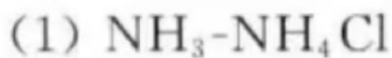


6-7 欲配制 $\text{pH}=9.5$ 的缓冲溶液, 可选用下列哪些缓冲溶液:

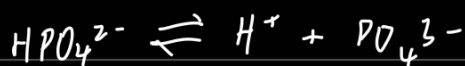
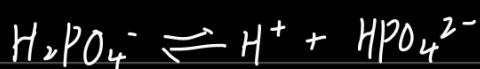
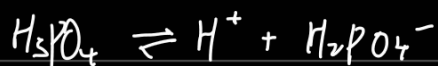


$\text{pH}=9.5$ 为碱性

(1) 与 (2) 可选

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6-27 $\text{H}_2\text{PO}_4^- \text{-HPO}_4^{2-}$ 是人体血液中重要的缓冲体系之一, 求 H_2PO_4^- 和 HPO_4^{2-} 浓度分别为 $0.050\text{mol} \cdot \text{L}^{-1}$ 和 $0.10\text{mol} \cdot \text{L}^{-1}$ 时的 pH 。



$$\text{pH} = \text{pK}_a + \log \left(\frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} \right)$$

$$\text{pK}_a = 7.2$$

$$\begin{aligned} \text{pK}_a &= -\log(K_a) \\ &= -\log(6.2 \times 10^{-8}) \\ &= 7.2 \end{aligned}$$

$$\text{pH} = 7.2 + \log \left(\frac{0.1}{0.05} \right) = 7.2 + \log 2 = 7.5$$

6-40 在 $[\text{Ag}(\text{NH}_3)_2]^+$ 的氨溶液中, 已知 $[\text{Ag}(\text{NH}_3)_2]^+$ 的浓度为 $0.10\text{mol} \cdot \text{L}^{-1}$, NH_3 的浓度为 $1.0\text{mol} \cdot \text{L}^{-1}$ 。若在此溶液中加入 $\text{Na}_2\text{S}_2\text{O}_3(\text{s})$, 使 $\text{Na}_2\text{S}_2\text{O}_3$ 初始浓度为 $2.2\text{mol} \cdot \text{L}^{-1}$ 。(假设加入 $\text{Na}_2\text{S}_2\text{O}_3$ 后溶液的体积不变)。求平衡时溶液中 $[\text{Ag}(\text{NH}_3)_2]^+$, $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, NH_3 和 $\text{S}_2\text{O}_3^{2-}$ 的浓度(已知 $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ 的 $K_{\text{不稳}}^\ominus = 3.4 \times 10^{-14}$ 。)



反应前	0.1	2.2	0	1.0
变化量	0.1-x	2(0.1-x)	0.1-x	2(0.1-x)
反应后	x	2.2-2(0.1-x)	0.1-x	1.0+2(0.1-x)

$$\text{总反应平衡常数 } K^\ominus = \frac{K_{\text{不稳}}^\ominus [Ag(NH_3)_2]^+}{K_{\text{稳}}^\ominus [Ag(S_2O_3)_2]^{3-}} = \frac{10^{-7.05}}{3.4 \times 10^{-14}} = 2.62 \times 10^6$$

$$2.2 - 2(0.1 - x) \approx 2.0 \quad 0.1 - x \approx 0.1 \quad 1.0 + 2(0.1 - x) \approx 1.2$$

$$K^\ominus = \frac{(0.1-x)(1.2-2x)^2}{x(2+2x)^2} = 2.62 \times 10^6 \approx \frac{0.1(1.2)^2}{x(2)^2} = 2.62 \times 10^6$$

$$x = 1.374 \text{ mol} \cdot \text{L}^{-1}$$

$$c(NH_3) = 1.2 \text{ mol} \cdot \text{L}^{-1} \quad c(S_2O_3^{2-}) = 2.0 \text{ mol} \cdot \text{L}^{-1}$$

$$c([Ag(NH_3)_2]^+) = 1.374 \times 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

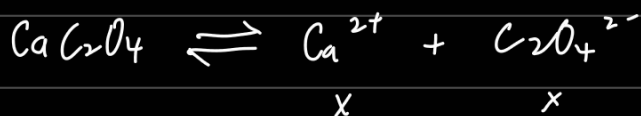
$$c([Ag(S_2O_3)_2]^{3-}) = 0.1 \text{ mol} \cdot \text{L}^{-1}$$

6-41 假设溶液的体积近似于溶剂水的体积,求下列难溶化合物在水中的溶解度 ($\text{mol} \cdot \text{L}^{-1}$) 和溶解度 ($\text{g}/100\text{g}$ 水):

(1) CaC_2O_4 ; (2) Ag_2CrO_4 ; (3) $\text{Ca}_3(\text{PO}_4)_2$ 。

$$(1) \text{CaC}_2\text{O}_4 \quad K_{\text{sp}}^\ominus = 4 \times 10^{-9} \quad M(\text{CaC}_2\text{O}_4) = 128.09$$

$$\text{平衡时, } K_{\text{sp}}^\ominus = [c(\text{Ca}^{2+})] \cdot [c(\text{C}_2\text{O}_4^{2-})] = 4 \times 10^{-9} \text{ mol} \cdot \text{L}^{-1}$$



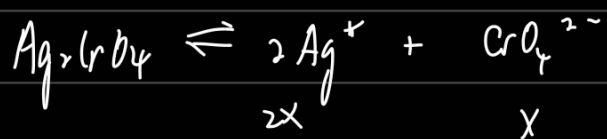
$$x \cdot x = 4 \times 10^{-9} \quad x = 6.325 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{CaC}_2\text{O}_4 \text{ 的溶解度 } (\text{mol} \cdot \text{L}^{-1}) \text{ 为 } 6.325 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{CaC}_2\text{O}_4 \text{ 的溶解度 } (\text{g}/100\text{g 水}) \text{ 为 } c(\text{Ca}^{2+}) \times M(\text{CaC}_2\text{O}_4) \times V(\text{溶液})$$

$$= 0.81 \text{ mg}/100 \text{ g 水}$$

(2) Ag_2CrO_4 $K_{sp}^\theta = 1.1 \times 10^{-12}$ $M(\text{Ag}_2\text{CrO}_4) = 331.7301$



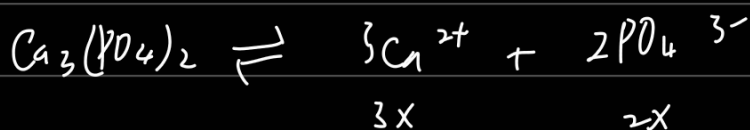
平衡时, $K_{sp}^\theta = [\text{c}(\text{Ag}^+)]^2 [\text{c}(\text{CrO}_4^{2-})] = 1.1 \times 10^{-12}$

$$4x^2 \cdot x = 1.1 \times 10^{-12} \quad x = 6.503 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

Ag_2CrO_4 的溶解度为 $6.503 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$

$$\begin{aligned} \text{Ag}_2\text{CrO}_4 \text{ 的溶解度 (g/100g 水) 为 } & \text{c}(\text{CrO}_4^{2-}) \times M(\text{Ag}_2\text{CrO}_4) \times V(\text{溶液}) \\ & = 6.503 \times 10^{-5} \times 331.7301 \times 0.1 \\ & = 2.16 \text{ mg/100g 水} \end{aligned}$$

(3) $\text{Ca}_3(\text{PO}_4)_2$ $K_{sp}^\theta = 2.0 \times 10^{-29}$ $M(\text{Ca}_3(\text{PO}_4)_2) = 310.177$



平衡时, $K_{sp}^\theta = [\text{c}(\text{Ca}^{2+})]^3 [\text{c}(\text{PO}_4^{3-})]^2 = 2.0 \times 10^{-29} \text{ mol} \cdot \text{L}^{-1}$

$$(3x)^3 (2x)^2 = 2.0 \times 10^{-29} \quad x = 7.137 \times 10^{-7} \text{ mol} \cdot \text{L}^{-1}$$

$\text{Ca}_3(\text{PO}_4)_2$ 的溶解度 ($\text{mol} \cdot \text{L}^{-1}$) 为 $7.137 \times 10^{-7} \text{ mol} \cdot \text{L}^{-1}$

$$\begin{aligned} \text{Ca}_3(\text{PO}_4)_2 \text{ 的溶解度 (g/100g 水) 为 } & \frac{1}{3} \text{c}(\text{Ca}^{2+}) \times M(\text{Ca}_3(\text{PO}_4)_2) \times V(\text{溶液}) \\ & = 0.022 \text{ mg/100g 水} \end{aligned}$$

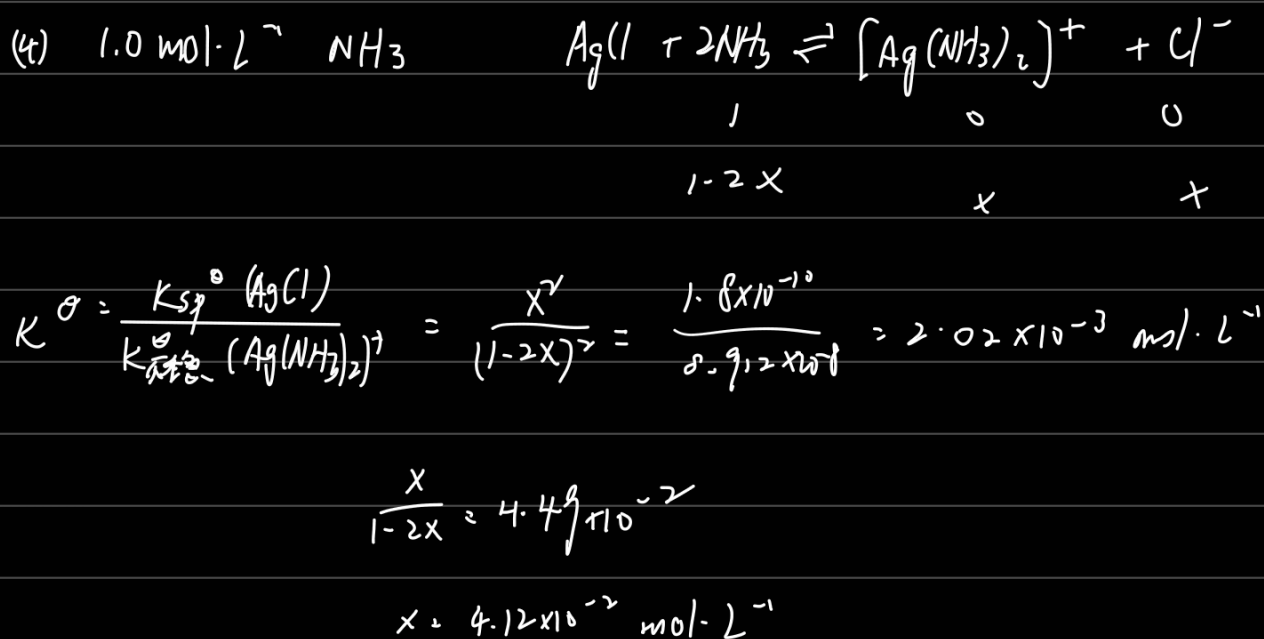
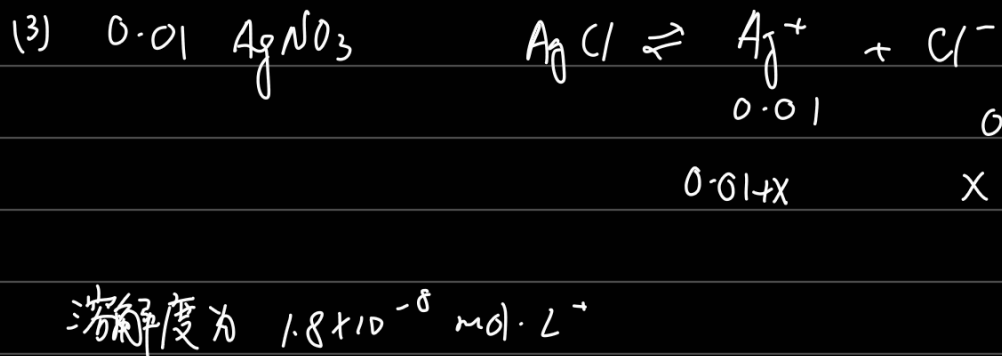
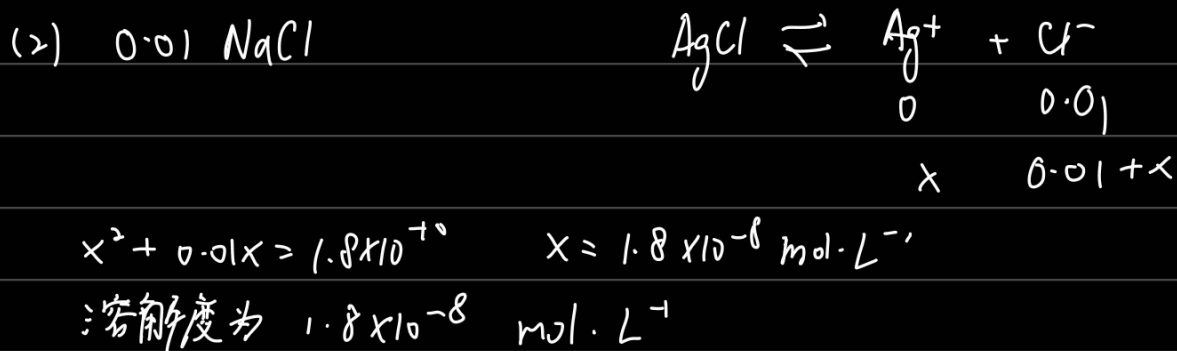
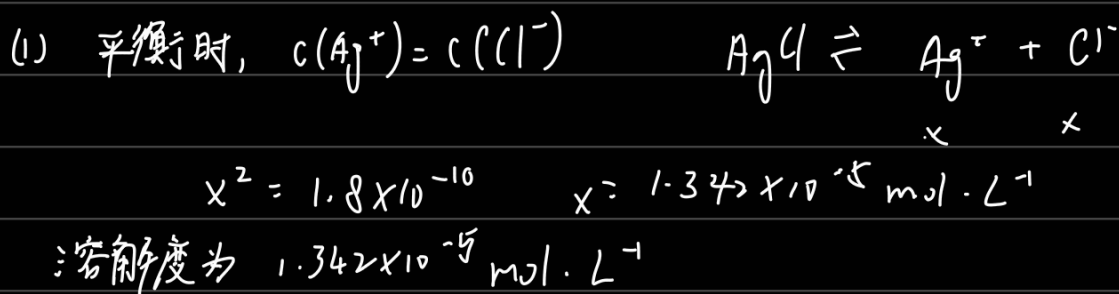
6-43 计算在下列溶液中 AgCl 的溶解度:

(1) 在纯水中;

(2) 在 $0.010 \text{ mol} \cdot \text{L}^{-1}$ NaCl 溶液中;

(3) 在 $0.010 \text{ mol} \cdot \text{L}^{-1}$ AgNO₃ 溶液中;

(4) 在 $1.0 \text{ mol} \cdot \text{L}^{-1}$ NH₃ 溶液中.



6-45 在含有 Cl^- , Br^- 和 I^- 等离子的混合溶液中, 各离子浓度均为 $0.10 \text{ mol} \cdot \text{L}^{-1}$, 若向混合溶液中逐滴加入 AgNO_3 溶液, 问哪种离子先沉淀? 当最后一种离子开始沉淀时, 其余两种离子是否已沉淀完全?

$$K_{\text{sp}}^\ominus(\text{AgCl}) = 1.8 \times 10^{-10} \quad K_{\text{sp}}^\ominus(\text{AgI}) = 8.7 \times 10^{-17} \quad K_{\text{sp}}^\ominus(\text{AgBr}) = 5 \times 10^{-13}$$

$$K_{\text{sp}}^\ominus(\text{AgI}) \text{ 最小} \rightarrow \text{I}^- \text{ 最先沉淀}$$

$$K_{\text{sp}}^\ominus(\text{AgCl}) = c(\text{Ag}^+) \cdot c(\text{Cl}^-) \quad c(\text{Ag}^+) = \frac{K_{\text{sp}}^\ominus(\text{AgCl})}{0.1} = 1.8 \times 10^{-9} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{此时, } c(\text{Br}^-) = \frac{K_{\text{sp}}^\ominus(\text{AgBr})}{c(\text{Ag}^+)} = 2.778 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} > 1 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \Rightarrow \text{未沉淀完全}$$

$$c(\text{I}^-) = \frac{K_{\text{sp}}^\ominus(\text{AgI})}{c(\text{Ag}^+)} = 4.611 \times 10^{-8} \text{ mol} \cdot \text{L}^{-1} < 1 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \Rightarrow \text{沉淀完全}$$

7-1 说明下列符号及术语的意思:

$E^\ominus$, E , $E_{\text{电池}}^\ominus$, $E_{\text{电池}}$, F , 正极和负极以及阴极和阳极。

$E^\ominus$ 是标准状态下电池的电极电势

E 是任意状态下电池的电极电势

$E_{\text{电池}}^\ominus$ 是标准状态下电池的电动势

$E_{\text{电池}}$ 是任意状态下电池的电动势

F 是法拉第常数 $F = 96485 \text{ C} \cdot \text{mol}^{-1} = 96485 \text{ J} \cdot \text{V}^{-1} \cdot \text{mol}^{-1}$

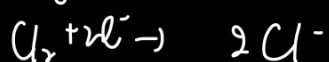
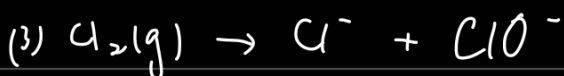
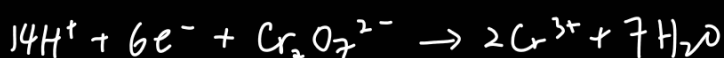
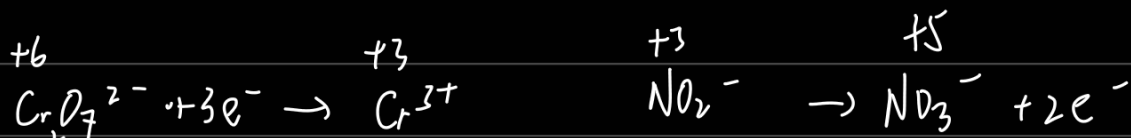
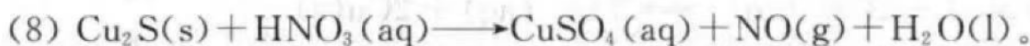
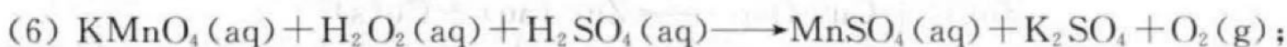
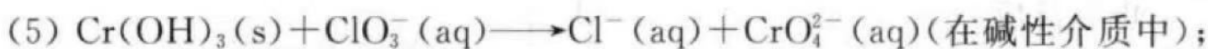
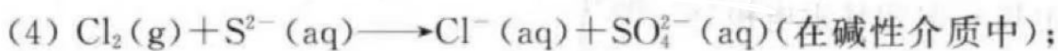
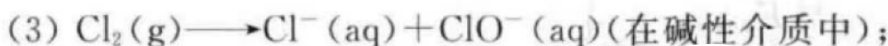
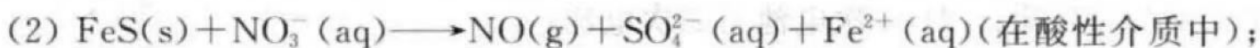
发生氧化反应的阳极, 发生还原反应的是阴极

电势高的(电子流入的)是正极, 电势低的(电子流出的)是负极

7-8 电化学电池中, 是否一定需要用盐桥或多孔板将两半电池隔开? 举一例。

不一定。锂离子电池使用的是固态电解质, 因此不需要隔开。

7-21 用离子-电子法配平下列氧化还原反应方程式：



C

